

Math 140 Spring 2023 - Worksheet 3.6

Section:

Name:

Try everything by hand.

1. Find the derivative of y with respect to x .

(a) $e^{xy} = 2x + y$.

(b) $\ln(x - y) = xy$.

2. Assume that x, y are differentiable functions of t . Find dy/dt in terms of x, y , and dx/dt for the function $x \sin y = 2$.

3. Find an equation of the tangent line to the graph of the equation at the given point. (Hint: You need the derivative of y with respect to x at the given point.)

$$xy^2 = 1, \quad (2, -3).$$